Database Systems
Practice Examination

Preamble

Please read the following instructions carefully:

- This exam has seven questions and 13 pages. Make sure your copy contains all exercises and all questions. The following table summarizes the questions in this exam and their weights.

Question	**Points**
1. Relational Algebra	10
2. SQL	20
3. ER Modeling	16
4. Design Theory	14
5. Physical Design	12
6. Query Processing and Optimization	14
7. Transactions, Concurrency Control, and Recovery	14
Total	**100**

- Read the text of each exercise carefully before solving it.
- If you get stuck on an exercise, proceed with another one and come back later.
- If your solution is based on an assumption that you have made, please add a comment for clarification.
- Aids are allowed during the exam, i.e., you can consult aids such as books, notes, etc.
- As a reference, we recommend the course material (book, slides, exercises, etc.) as it complies with the notation used in the exam.
- It is NOT allowed to seek help from or consult others (human beings, artificial intelligence, etc.) during the exam.
- Good luck!

1 Relational Algebra (10 points)

Consider the following relations capturing information about a movie streaming service:

```
studio(sid, studioName, country, foundedYear)
movie(mid, title, genre, releaseYear, sid -> studio)
subscriber(uid, username, email, subscription)
viewing(uid -> subscriber, mid -> movie, watchDate, rating)
```

where the primary keys are underlined and the foreign keys are denoted by arrows.

Fill in the missing information that should go into the boxes to make the query compute the required information. Note that your solution should not output additional information.

To enter the operation symbols into the answer sheet, please use the following replacements:

Operation symbol	**What to write in the answer sheet**
ρ	rho
γ	gamma
Π	pi
σ	sigma
$\bowtie$	join
$\times$	x
$-$	minus

1.1 (4 points)

Find all subscribers who have watched movies from studios in 'USA' and gave a rating of 5. For each such subscriber, list their uid and username.

$$\boxed{\text{Box 1}}_{\text{uid, username}} (\boxed{\text{Box 2}}_{\text{rating} = 5} (\text{subscriber} \bowtie \text{viewing} \bowtie \text{movie} \bowtie (\boxed{\text{Box 3}}_{\text{country} = \text{'USA'}} (\boxed{\text{Box 4}}))))$$

1.2 (6 points)

Find all subscribers who have NOT watched any Horror movie. For each such subscriber, list their uid.

$$\Pi_{\text{uid}}(\boxed{\text{Box 5}})\ \boxed{\text{Box 6}}\ \Pi_{\text{uid}}(\text{viewing} \bowtie_{\text{viewing.mid} = \text{movie.mid}} (\boxed{\text{Box 7}}_{\text{genre} = \text{'Horror'}} (\boxed{\text{Box 8}})))$$

2 SQL (20 points)

2.1 Evaluate whether the following statements about SQL are true or false.

2.1.1 (2 points)

The expression `NULL = NULL` evaluates to TRUE in SQL.

(a) True (b) False

2.1.2 (2 points)

The condition `x IN (empty set)` evaluates to FALSE, while `x > ALL (empty set)` evaluates to TRUE.

(a) True (b) False

2.2 Consider again the relational schema from Question 1:

```
studio(sid, studioName, country, foundedYear)
movie(mid, title, genre, releaseYear, sid -> studio)
subscriber(uid, username, email, subscription)
viewing(uid -> subscriber, mid -> movie, watchDate, rating)
```

where the primary keys are underlined and the foreign keys are denoted by arrows.

Answer the following questions:

2.2.1 (4 points)

Which of the following queries correctly finds the average rating per movie genre? For each genre, list the genre and the average. Choose **only one** option:

(a) SELECT genre, AVG(rating)
FROM viewing, movie
GROUP BY genre;

(b) SELECT m.genre, AVG(v.rating)
FROM viewing v JOIN movie m ON v.mid = m.mid
GROUP BY m.genre;

(c) SELECT genre, AVG(rating)
FROM viewing
GROUP BY genre;

(d) SELECT AVG(rating)
FROM viewing, movie
WHERE genre GROUP BY genre;

(e) Both (a) and (b) are correct.

(f) None of the above

2.2.2 (4 points)

Which of the following queries correctly finds subscribers who have watched more than 10 movies? For each such subscriber, list their uid and the count. Choose **only one** option:

(a) SELECT uid, COUNT(*)
FROM viewing
WHERE COUNT(*) > 10
GROUP BY uid;

(b) SELECT uid, COUNT(*)
FROM viewing
GROUP BY uid
HAVING COUNT(*) > 10;

(c) SELECT uid
FROM viewing
GROUP BY uid
HAVING mid > 10;

(d) SELECT uid
FROM viewing
HAVING COUNT(*) > 10;

(e) Both (a) and (b) are correct.

(f) None of the above

2.2.3 (4 points)

Fill in the missing information that should go into the boxes of the following query: find all studios that have at least one Action movie. For each such studio, list sid and studioName.

```
SELECT DISTINCT s.sid, s.studioName
FROM studio s
WHERE [Box 1] (
    SELECT * FROM movie m
    WHERE m.sid = s.sid [Box 2] m.genre = 'Action'
);
```

2.2.4 (4 points)

Consider the following query that finds subscribers who have never watched any movie:

```
SELECT s.uid, s.username
FROM subscriber s LEFT JOIN viewing v ON s.uid = v.uid
WHERE v.uid IS NULL;
```

Which of the following statements is **correct**? Choose **one or more** options:

(a) This query uses an outer join to preserve all subscribers.

(b) The WHERE clause filters for subscribers with no matching viewing records.

(c) This query is equivalent to: `SELECT uid, username FROM subscriber WHERE uid NOT IN (SELECT uid FROM viewing)`.

(d) If we replace LEFT JOIN with INNER JOIN, the query would return an empty result set (no rows).

(e) The condition `v.uid IS NULL` checks for subscribers without any viewings.

(f) All of the above.

3 ER Modeling (16 points)

3.1 Evaluate whether the following statements about ER model concepts are true or false.

3.1.1 (2 points)

A double ellipse in an ER diagram represents a multi-valued attribute.

(a) True (b) False

3.1.2 (2 points)

A weak entity type can be uniquely identified by its own attributes alone, without needing its owner entity's key.

(a) True (b) False

3.1.3 (2 points)

In a M:N (many-to-many) relationship, a separate relationship table must be created during ER-to-relational mapping.

(a) True (b) False

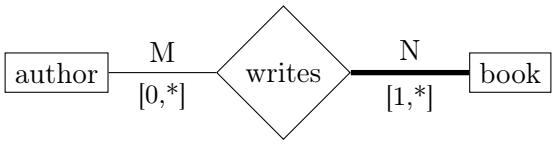
3.2 (4 points)

Consider the relationship *writes* where

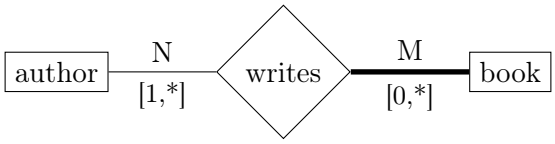
- each *book* must be written by at least one author (total participation)
- each *author* can write many books or no books (partial participation)

Which of the following ER diagram representations is **correct**? Choose **only one** option:

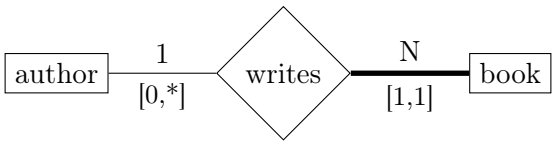
Note: Chen notation (1, N, M) is placed near the relationship diamond. [min, max] notation is placed near the entity. Double lines indicate total participation.



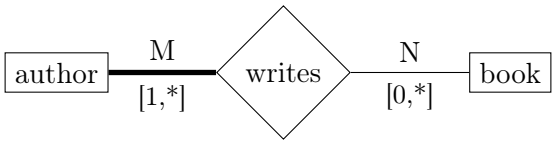
(a)



(b)



(c)

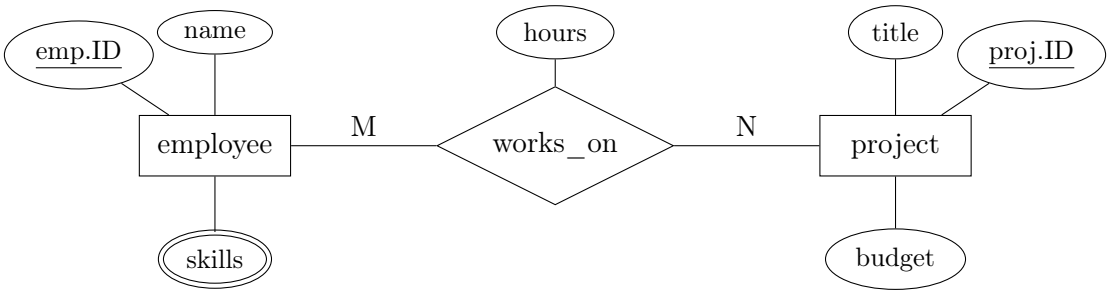


(d)

(e) None of the above

3.3 (6 points)

Consider the following ER diagram showing the relationship *works_on* between *employee* and *project*:



Note: The double ellipse around "skills" indicates a multi-valued attribute. The relationship is M:N (many employees work on many projects). The relationship has attribute "hours".

Which combination of relations should be created for this ER diagram? Note that we aim for a minimal schema. Choose **one or more** options:

(a) employee(emp.ID, name, skills)

(b) employee(emp.ID, name)

(c) skills(emp.ID → employee, skill)

(d) project(proj.ID, title, budget)

(e) works_on(emp.ID → employee, proj.ID → project, hours)

(f) works_on(emp.ID → employee, proj.ID → project, hours)

(g) project(proj.ID, title, budget, emp.ID → employee, hours)

(h) None of the above

4 Design Theory (14 points)

4.1 Evaluate whether the following statements about design theory are true or false.

4.1.1 (2 points)

The closure of an attribute set includes all attributes that can be functionally determined by it.

(a) True (b) False

4.1.2 (2 points)

A relation is in BCNF if and only if it is in 3NF.

(a) True (b) False

4.2 Consider a relation schema $R = (P, Q, R, S, T)$ with the following functional dependencies (FDs):

$$\begin{aligned} PQ &\rightarrow R \\ R &\rightarrow ST \\ S &\rightarrow Q \end{aligned}$$

Answer the following questions:

4.2.1 (4 points)

Which of the following attributes (sets) can be a **superkey** of R? Choose **one or more** options:

(a) P

(b) Q

(c) R

(d) S

(e) PQ

(f) PR

(g) PS

(h) RS

(i) PQR

(j) PQRST

(k) None of the above

4.2.2 (3 points)

Which of the following statements is correct about the normalization status of relation R? Choose **only one** option:

(a) The relation R is in BCNF and 3NF.

(b) The relation R is in BCNF but not in 3NF.

(c) The relation R is in 3NF but not in BCNF.

(d) The relation R is not in BCNF or 3NF.

(e) None of the above

4.2.3 (3 points)

Which of the following decompositions of the relational schema $R = (P, Q, R, S, T)$ is **lossless**? Choose **one or more** options:

(a) $R_1 = (P, Q, R)$ and $R_2 = (P, S, T)$

(b) $R_1 = (P, Q, R)$ and $R_2 = (R, S, T)$

(c) $R_1 = (P, R, S)$ and $R_2 = (Q, R, T)$

(d) $R_1 = (S, Q, R)$ and $R_2 = (P, Q, T)$

(e) None of the above

5 Physical Design (12 points)

5.1 Evaluate whether the following statements about B+ Trees are true or false.

5.1.1 (2 points)

In a B+ Tree, internal nodes contain copies of keys that also appear in leaf nodes.

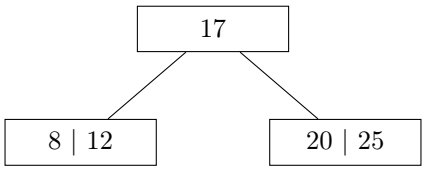
(a) True (b) False

5.1.2 (2 points)

For a B+ Tree with degree $d = 4$, a leaf node can contain a maximum of 3 keys.

(a) True (b) False

5.2 Consider the following B+ Tree with degree $d = 3$.



Visual representation. Actual leaf structure:

- Leaves: [6* 8*] | [12* 15*] | [17* 20*] | [25* 30*]
- Root: contains key 17 with 2 pointers
- Internal nodes: Left child contains keys [8, 12]; Right child contains keys [20, 25]

Please make the following assumptions:

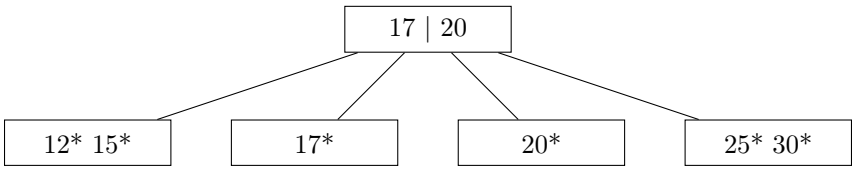
- The left pointer of a key k in a non-leaf node leads towards keys less than k, while the right pointer leads towards keys greater than or equal to k.
- A leaf node underflows when the number of keys goes below $\lceil \frac{d-1}{2} \rceil$.
- An internal node underflows when the number of pointers goes below $\lceil \frac{d}{2} \rceil$.

Answer the following questions:

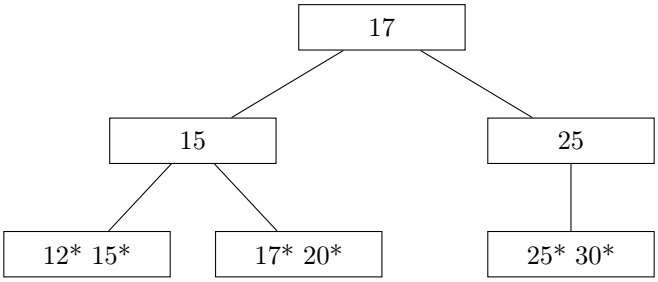
5.2.1 (4 points)

Delete 6*, then delete 8*. What is the resulting tree? Choose **only one** option:

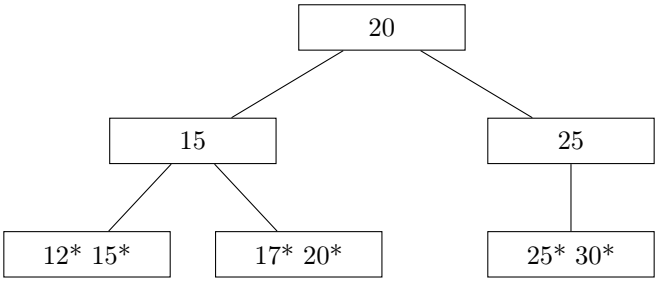
(a)



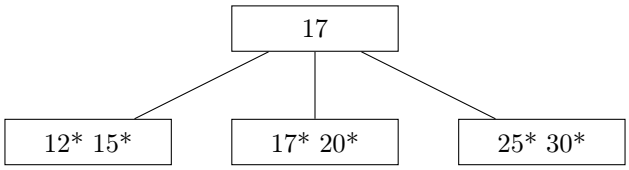
(b)



(c)



(d)

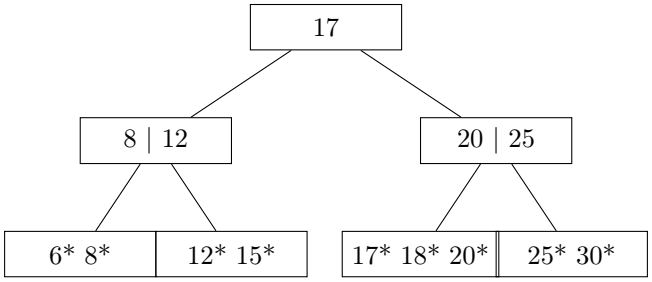


(e) None of the above

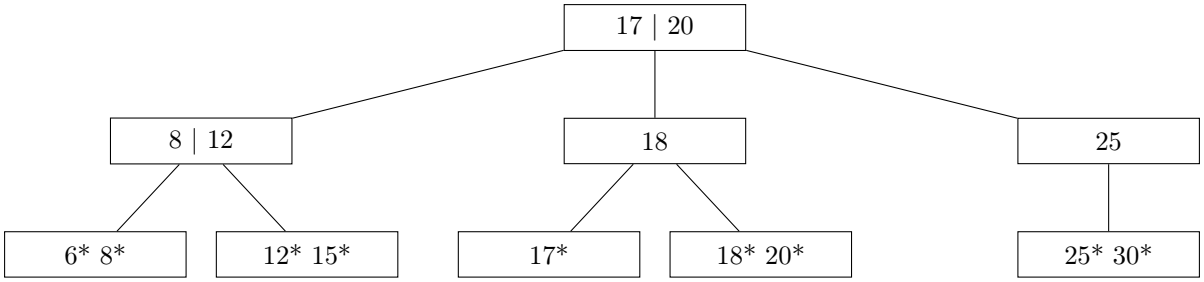
5.2.2 (4 points)

Consider the original tree (ignore operations in 5.2.1). Insert 18* into the tree. What is the resulting tree? Choose **only one** option:

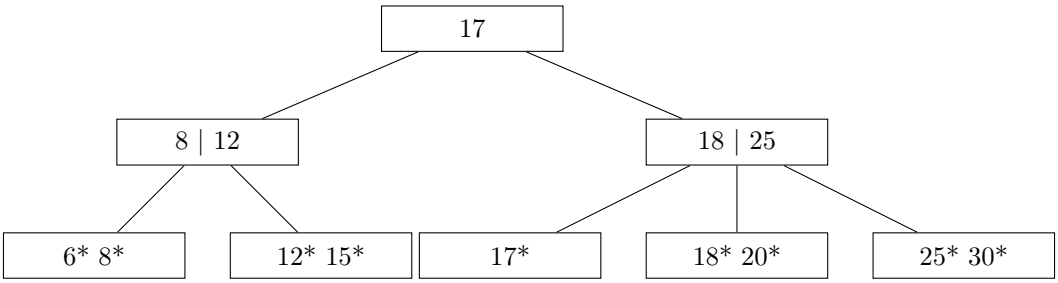
(a)



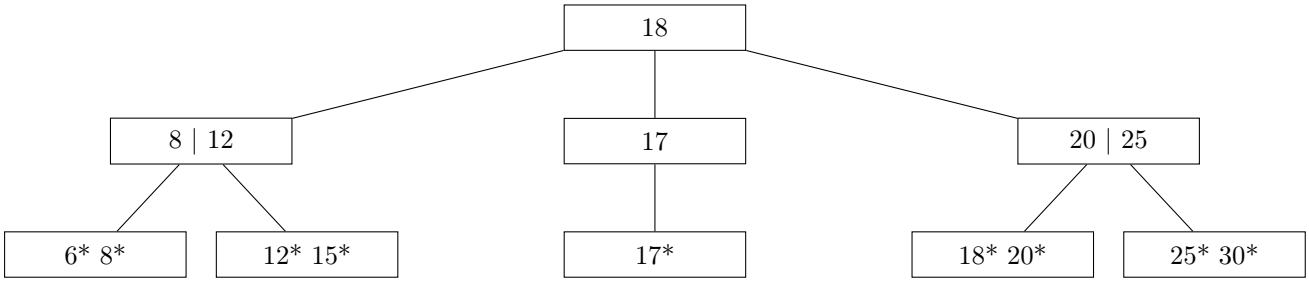
(b)



(c)



(d)



(e) None of the above

6 Query Processing and Optimization (14 points)

Consider a database containing relations s_1 and s_2 with the following properties: s_1 has 60,000 tuples, s_2 has 300,000 tuples, 30 tuples of s_1 fit on one page, and 15 tuples of s_2 fit on one page. Assume the computer has 250 buffer pages.

Answer the following questions:

6.1 (2 points)

Suppose you need to sort s_1 using the 2-Way External Merge Sort algorithm. How many passes are needed in Phase 2? Choose **only one** option:

(a) $\lceil \log_2 60,000 \rceil$

(b) $\lceil \log_2 2,000 \rceil$

(c) $\lceil \log_{249} \lceil 2,000/250 \rceil \rceil$

(d) $\lceil \log_{249} \lceil 60,000/250 \rceil \rceil$

(e) None of the above

6.2 (4 points)

Suppose you need to sort s_2 using the Multi-Way External Merge Sort algorithm. Which of the following statements is **correct**? Choose **one or more** options:

(a) The number of pages for s_2 is 20,000.

(b) The I/O cost in Phase 1 is 40,000 page transfers.

(c) The number of initial sorted runs created in Phase 1 is $\lceil 20,000/250 \rceil = 80$.

(d) The number of passes needed in Phase 2 is $\lceil \log_{249} 80 \rceil = 1$.

(e) The total I/O cost for sorting is $2 \times 20,000 \times (1 + 1) = 80,000$ page transfers.

(f) None of the above

6.3 (4 points)

Suppose you need to join s_1 and s_2 using the block-oriented nested-loop join algorithm with s_1 as outer. Which of the following statements is **correct**? Choose **only one** option:

(a) The number of pages for s_1 is 2,000 and for s_2 is 20,000.

(b) The total I/O cost is $2,000 + 2,000 \times 20,000 = 40,002,000$ page transfers.

(c) If we use s_2 as outer instead, the cost would be $20,000 + 20,000 \times 2,000 = 40,020,000$ page transfers.

(d) Both (a) and (b) are correct.

(e) None of the above

6.4 (4 points)

Given a relation $s(X, Y, Z)$ with $n_s = 5000$, $V(X, s) = 200$, $V(Y, s) = 250$, and $V(Z, s) = 50$, where n_s is the number of tuples in relation s and $V(X, s)$ is the number of distinct values that appear in s for attribute X.

Assume the range of values for attribute X is [100, 300], for attribute Y is [0, 500], and for attribute Z is [50, 150]. The values are uniformly distributed.

Which of the following statements is **correct**? Choose **one or more** options:

(a) The size estimation of the selection operation $\sigma_{Z=100}(s)$ is 100.

(b) The size estimation of the selection operation $\sigma_{X=200}(s)$ is 25.

(c) The size estimation of the selection operation $\sigma_{Y<250}(s)$ is 2500.

(d) The size estimation of the selection operation $\sigma_{X<200}(s)$ is 2500.

(e) The size estimation of the selection operation $\sigma_{Z>100}(s)$ is 2500.

(f) None of the above

7 Transactions, Concurrency Control, and Recovery (14 points)

7.1 Evaluate whether the following statements are true or false.

7.1.1 (2 points)

The REDO phase in log-based recovery processes the log in forward direction (from beginning to end).

(a) True (b) False

7.1.2 (2 points)

Deadlock can occur when two or more transactions are waiting for locks held by each other.

(a) True (b) False

7.2 We consider three transactions T1, T2, and T3 consisting of the following actions:

```
T1: R(M) W(N) C
T2: W(M) R(P) W(P) C
T3: R(N) W(M) R(P) C
```

R(X) represents reading object X, W(X) represents writing object X, and C represents Commit.

Consider the following two schedules:

Schedule (1):

Transaction	Operations
T1	R(M) W(N) C
T2	W(M) R(P) W(P) C
T3	R(N) W(M) R(P) C

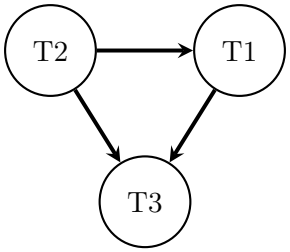
Schedule (2):

Transaction	Operations
T1	R(M) W(N) C
T2	W(M) R(P) W(P) C
T3	R(N) W(M) R(P) C

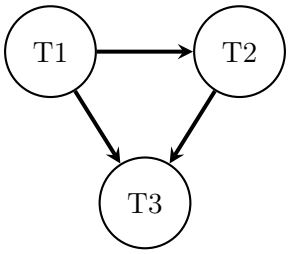
Answer the following questions:

7.2.1 (3 points)

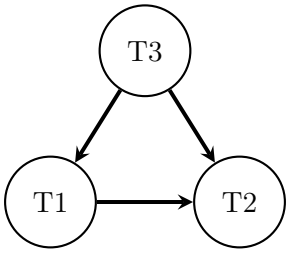
Which precedence graph of Schedule (1) is **correct**? Choose **one** option:



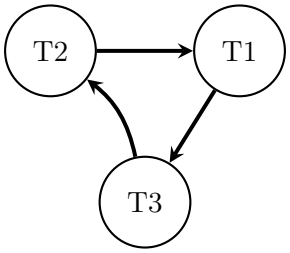
(a)



(b)



(c)



(d)

(e) None of the above

7.2.2 (4 points)

Which of the following statements about Schedule (1) is **correct**? Choose **one or more** options:

(a) Schedule (1) is a serial schedule.

(b) Schedule (1) is a conflict serializable schedule.

(c) Schedule (1) is a serializable schedule.

(d) The equivalent serial schedule of Schedule (1) can be T2, T1, T3.

(e) The equivalent serial schedule of Schedule (1) can be T1, T2, T3.

(f) None of the above

7.2.3 (3 points)

Analyze Schedule (2). Which type(s) of schedule does Schedule (2) belong to? Choose **one or more** options:

(a) Serial schedule

(b) Serializable schedule

(c) Conflict serializable schedule

(d) None of the above

ANSWERS SECTION

Question 1: Relational Algebra

1.1 (4 points)

Box 1: pi (projection)

Box 2: sigma (selection)

Box 3: sigma (selection)

Box 4: studio

Complete answer: $\Pi_{\text{uid, username}}(\sigma_{\text{rating}=5}(\text{subscriber} \bowtie \text{viewing} \bowtie \text{movie} \bowtie (\sigma_{\text{country} = \text{'USA'}}(\text{studio}))))$

Explanation: Join all tables to connect subscribers to studios through viewings and movies, then filter for USA studios with rating = 5, and project the required attributes.

1.2 (6 points)

Box 5: subscriber

Box 6: - (minus / set difference)

Box 7: sigma (selection)

Box 8: movie

Complete answer: $\Pi_{\text{uid}}(\text{subscriber}) - \Pi_{\text{uid}}(\text{viewing} \bowtie_{\text{viewing.mid} = \text{movie.mid}} (\sigma_{\text{genre} = \text{'Horror'}}(\text{movie})))$

Explanation: All subscriber UIDs MINUS those who have watched Horror movies = subscribers who have NOT watched any Horror movie. This is the standard "NOT" pattern in relational algebra.

Question 2: SQL

2.1.1 (2 points)

Answer: (b) False

Any comparison with NULL (including NULL = NULL) yields UNKNOWN, not TRUE. To check for NULL, you must use IS NULL or IS NOT NULL.

2.1.2 (2 points)

Answer: (a) True

x IN (empty set) is FALSE because x cannot be equal to any element in an empty set. x > ALL (empty set) is TRUE (vacuously true) because x is greater than all zero values in the empty set.

2.2.1 (4 points)

Answer: (b)

Option (b) correctly joins viewing and movie on mid and groups by genre. Option (a) is wrong because it creates a Cartesian product without proper join condition. Option (c) is wrong because viewing doesn't have a genre column. Option (d) has invalid syntax.

2.2.2 (4 points)

Answer: (b)

Correct: GROUP BY uid, then HAVING COUNT(*) > 10 filters groups. Cannot use aggregate functions like COUNT in WHERE clause (option a). Option (c) uses non-aggregated mid in HAVING. Option (d) is missing GROUP BY.

2.2.3 (4 points)

Box 1: EXISTS

Box 2: AND

Complete query:

```
SELECT DISTINCT s.sid, s.studioName
FROM studio s
WHERE EXISTS (
    SELECT * FROM movie m
    WHERE m.sid = s.sid AND m.genre = 'Action'
);
```

Explanation: EXISTS checks if the correlated subquery returns at least one row. For each studio, we check if there exists a movie record with matching sid and genre = 'Action'.

2.2.4 (4 points)

Answer: (f) All of the above

(a) TRUE: LEFT JOIN preserves all subscribers. (b) TRUE: WHERE v.uid IS NULL filters for those without viewings. (c) TRUE: equivalent to NOT IN approach. (d) TRUE: INNER JOIN only returns matching rows, but then filtering for IS NULL means no rows match (empty result). (e) TRUE: IS NULL checks for no matching records.

Question 3: ER Modeling

3.1.1 (2 points)

Answer: (a) True

A double ellipse in an ER diagram represents a multi-valued attribute - an attribute that can have multiple values for a single entity instance (e.g., phone numbers for an employee).

3.1.2 (2 points)

Answer: (b) False

A weak entity type CANNOT be uniquely identified by its own attributes alone. It requires the primary key of its owner (strong) entity combined with its partial key. For example, a Dependent entity needs Employee's PK + dependent name to be unique.

3.1.3 (2 points)

Answer: (a) True

In M:N relationships, neither entity can hold a FK to the other (would require multi-valued FKs). Therefore, a separate relationship table MUST be created with composite PK consisting of both entity PKs as FKs.

3.2 (4 points)

Answer: (a)

Interpretation of the requirements:

- Book [1,*]: must be written (min=1, total participation shown by double line) by at least one author (max=*)
- Author [0,*]: may write none (min=0, partial participation) or many books (max=*)
- Chen notation: M near author, N near book (M:N relationship - many authors can write many books)

Why other options are wrong:

- (b): [min,max] values swapped - would mean books have partial participation and authors have total
- (c): Shows 1:N relationship with wrong participation (each book by exactly one author)
- (d): [min,max] values completely swapped between entities

3.3 (6 points)

Answer: (b), (c), (d), (e)

Correct mapping (4 relations):

- **(b)** employee(<u>emp.ID</u>, name)
  - Strong entity with simple attributes only
  - skills is multi-valued → separate table (NOT included here)
- **(c)** skills(<u>emp.ID → employee, skill</u>)
  - Multi-valued attribute creates separate table
  - Composite PK: emp.ID + skill (both underlined)
- **(d)** project(<u>proj.ID</u>, title, budget)

  - Strong entity with all its simple attributes

- **(e)** works_on(emp.ID → employee, proj.ID → project, hours)
  - M:N relationship creates separate table
  - Composite PK: both FKs (emp.ID + proj.ID)
  - Relationship attribute hours goes in this table

Why other options are wrong:

- (a): skills should NOT be in employee table (it's multi-valued)
- (f): Missing underlines for composite PK (both FKs must be part of PK in M:N)
- (g): Cannot add FK to project for M:N (would need multiple values)

Question 4: Design Theory

4.1.1 (2 points)

Answer: (a) True

The closure α^+ contains all attributes functionally determined by α.

4.1.2 (2 points)

Answer: (b) False

BCNF is stricter than 3NF. All BCNF relations are in 3NF, but not all 3NF relations are in BCNF.

4.2.1 (4 points)

Answer: (e), (f), (g), (j)

Computing closures:

- $P^+ = \{P\}$ - not a superkey
- $(PQ)^+ = \{P, Q, R, S, T\}$ (using $PQ \to R$, then $R \to ST$) - **superkey!**
- $(PR)^+ = \{P, R, S, T, Q\}$ (using $R \to ST$, $S \to Q$) - **superkey!**
- $(PS)^+ = \{P, S, Q, R, T\}$ (using $S \to Q$, $PQ \to R$, $R \to ST$) - **superkey!**
- All supersets of these are also superkeys.

4.2.2 (3 points)

Answer: (c)

Check BCNF: For $S \to Q$, S is not a superkey ($S^+ = \{S, Q\}$). NOT in BCNF.

Check 3NF: For $S \to Q$, Q is part of candidate key PQ, so it's acceptable in 3NF. IN 3NF.

4.2.3 (3 points)

Answer: (b)

For lossless: $(R_1 \cap R_2)^+$ must contain R_1 or R_2.

(b) $R_1 = (P, Q, R)$, $R_2 = (R, S, T)$. Common: $\{R\}$. $R^+ = \{R, S, T, Q\}$. Contains R_2! **Lossless!**

Question 5: Physical Design

5.1.1 (2 points)

Answer: (a) True

In B+ trees, internal nodes guide the search but all actual data pointers/records are in leaves. Keys may be copied up to internal nodes.

5.1.2 (2 points)

Answer: (a) True

For degree $d = 4$: max keys in leaf $= d - 1 = 3$.

5.2.1 (4 points)

Answer: (b)

After deleting 6* from [6* 8*], we have [8*] (1 key). After deleting 8*, we have [] (0 keys, severe underflow). For $d = 3$: min keys $= \lceil (3-1)/2 \rceil = 1$.

Must merge with sibling [12* 15*] to get [12* 15*]. Then restructure the tree. The leaves become: [12* 15*] | [17* 20*] | [25* 30*] Internal structure adjusts. Root has one separator.

Result: Root [17], with internal nodes [15] on left and [25] on right.

5.2.2 (4 points)

Answer: (c)

Inserting 18* into [17* 20*] makes [17* 18* 20*] (3 keys). For $d = 3$: max keys in leaf $= 2$. This overflows!

Split: [17*] and [18* 20*]. Copy up 18. New structure: Root [17], Left internal [8|12], Right internal [18|25]

Question 6: Query Processing and Optimization

6.1 (2 points)

Answer: (b)

Pages for s_1: $60,000/30 = 2,000$. 2-Way sort Phase 2 passes: $\lceil \log_2 2,000 \rceil = 11$.

6.2 (4 points)

Answer: (a), (b), (c), (d), (e)

Pages for s_2: $300,000/15 = 20,000$. (a) **Correct!**

Phase 1 I/O: read + write $= 2 \times 20,000 = 40,000$. (b) **Correct!**

Initial runs: $\lceil 20,000/250 \rceil = 80$. (c) **Correct!**

Phase 2 passes: $\lceil \log_{249} 80 \rceil = 1$. (d) **Correct!**

Total I/O: Phase 1 (40,000) + Phase 2 (1 pass × 40,000) = 80,000. (e) **Correct!**

6.3 (4 points)

Answer: (d)

Both (a) and (b) are correct. Pages: s_1 = 2,000, s_2 = 20,000. Cost with s_1 as outer: $2,000 + 2,000 \times 20,000$.

6.4 (4 points)

Answer: (a), (b), (c), (d), (e)

(a) $\sigma_{Z=100}(s)$: $5000/50 = 100$. **True!**

(b) $\sigma_{X=200}(s)$: $5000/200 = 25$. **True!**

(c) $\sigma_{Y<250}(s)$: $5000 \times \frac{250-0}{500-0} = 2500$. **True!**

(d) $\sigma_{X<200}(s)$: $5000 \times \frac{200-100}{300-100} = 5000 \times 0.5 = 2500$. **True!**

(e) $\sigma_{Z>100}(s)$: $5000 \times \frac{150-100}{150-50} = 5000 \times 0.5 = 2500$. **True!**

Question 7: Transactions, Concurrency Control, and Recovery

7.1.1 (2 points)

Answer: (a) True

REDO processes log forward (beginning to end) to reapply committed changes.

7.1.2 (2 points)

Answer: (a) True

Deadlock occurs when transactions form a cycle in the wait-for graph, each waiting for locks held by others.

7.2.1 (3 points)

Answer: (a)

Conflicts in Schedule (1):

- T2: W(M) before T1: R(M) → T2 → T1
- T1: W(N) before T3: R(N) → T1 → T3
- T2: W(P) before T3: R(P) → T2 → T3

Graph: T2 → T1, T1 → T3, T2 → T3

7.2.2 (4 points)

Answer: (b), (c), (d)

(b) Acyclic precedence graph means conflict serializable. **True!**

(c) All conflict serializable schedules are serializable. **True!**

(d) Topological sort gives: T2, T1, T3. **True!**

7.2.3 (3 points)

Answer: (b), (c)

Conflicts in Schedule (2):

- T1: R(M) before T2: W(M) → T1 → T2
- T2: W(M) before T3: W(M) → T2 → T3
- T1: W(N) before T3: R(N) → T1 → T3
- T2: W(P) before T3: R(P) → T2 → T3

The precedence graph is: T1 → T2, T2 → T3, T1 → T3.

This is acyclic! So it IS conflict serializable and serializable. Equivalent serial order: T1, T2, T3.